MCBIOS 2026

PROGRAM BOOK

Bridging Data, AI, and Innovation to Transform Health

The 22st Annual Meeting of the MidSouth Computational Biology and Bioinformatics Society (MCBIOS)

Moffitt Cancer Center
Tampa, FL
March 27-29, 2026

MCBIOS.COM

MIDSOUTH COMPUTATIONAL BIOLOGY &
BIOINFORMATICS SOCIETY

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Qianqian Song, Ph.D.	*University of Florida*
Xuefeng Wang, Ph.D.	*Moffitt Cancer Center*

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Acknowledgements to MCBIOS 2026 Sponsors

MCBIOS 2026 Program At A Glance

DAY 1 March 27, 2026 (Friday)			
7:30 am - 8:30 am	REGISTRATION & BREAKFAST		
	Ted & Marty Couch Auditorium	SRB Ferman Family Room	SRB David Murphey Room
8:30 - 8:45 am	Welcome Remarks		
8:45 - 9:45 am	KEYNOTE 1 Rafael A. Irizarry – Dana-Farber		
9:45 - 11:00 am	SESSION 101 Chair Shenglin Mei Innovative Computational Approaches for Decoding Biological Complexity from Single-Cell Data	SESSION 102 Chairs Jianrong Wang Multi-omics integration to decipher disease mechanisms	SESSION 103 Chair Yan Guo AI in Genomics
11:00 – 11:15 am	Break		
11:15 – 12:30 pm	SESSION 201 Chair Jing Su Computational Digital Twins for Translational Biomedical Informatics	SESSION 202 Chairs Xijin Ge Timothy Shaw Next-Generation Web Visual Analytics for Bioinformatics and Artificial Intelligence	SESSION 203 Chair Zongliang Yue From Omics to Network Pharmacology and Systems Medicine
12:30 – 12:40 pm	Group Pictures		
12:40 – 1:30 pm	LUNCH		
1:30 – 2:30 pm	KEYNOTE 2 Ting Wang		
2:30 – 3:45 pm	SESSION 301 Chairs Dung-Tsa Chen Aik Choon Tan Adverse Event: A Gold Mine for Data Science in Cancer Research	SESSION 302 Chairs Jatinder Lamba Timothy I Shaw Next-Generation Genomics and Bioinformatics for Pediatric Cancer and Congenital Disorders	SESSION 303 Chair Hui Jiang From Data to Discovery: Statistical Learning in Biomedicine
3:45 -4:00 pm	Break		

4:00 - 5:15 pm	Session 304 Chair Zhaotong Lin Single-Cell and Multi-Omics Genomics: Bridging Data and Innovation to Transform Health	SESSION 305 Chairs Leon Hardy Timothy I Shaw Student Flash Talks from the ISCB Regional Meeting 2025	Session 306 YSEA (Post-doc / Undergrad) YSEA (Grad Student)
5:15 - 6:30 pm	MCBIOS Board Member Voting (Auditorium)		
	RECEPTION, POSTER SESSION (SRB Atrium)		
6:30 PM	MCBIOS Business Session (By Invitation Only)		

DAY 2 March 28, 2026 (Saturday)			
7:30 - 8:30 am	BREAKFAST		
	Ted & Marty Couch Auditorium	SRB Ferman Family Room	SRB David Murphey Room
8:30 - 9:30 am	KEYNOTE 3 Issam El Naqa		
9:30 - 9:45 am	Break		
9:45 - 11:00 am	SESSION 401 Chairs Zhana Duren Xiuwei Zhang Peng Qiu Advanced Computational Methods in Single-cell and Spatial Omics	SESSION 402 Chair Li Chen Artificial Intelligence Innovations for Spatial Transcriptome Analysis	SESSION 403 Chair Timothy I Shaw Careers in Bioinformatics: A Conversation with Industry Leaders
11:00 - 11:15 am	Break		
11:15 - 12:30 pm	SESSION 501 Chair Sophia Song Mingxiang Teng AI in precision medicine	SESSION 502 Chair Zhaohui “Steve” Qin Novel machine learning methods in genetics and genomics	SESSION 503 Chair Oluwatosin Oluwadare AI-Driven 3D Genome Reconstruction and Multimodal Learning
12:30 -1:30 pm	LUNCH		
1:30 – 2:30 pm	KEYNOTE 4 Jinghui Zhang		
2:30 - 3:45 pm	SESSION 601 Chair Ece Uzun	Session 602 Chair Qianqian Song	SESSION 603 Chairs Inimary Toby-Ogundeji

	Genomics and Translational Bioinformatics Working Group Session: AI Applications in Genomics and Bioinformatics	AI and Foundation Models in Biomedicine	Emil Jivishov AI-Powered Bio-Discovery: A Multi-Level Collaborative Codeathon
3:45 - 4:00 pm	Break		
4:00 - 5:15 pm	SESSION 701 Chairs Peng Qiu Xiuwei Zhang Zhana Duren Machine Learning Advances in Single-cell Omics	SESSION 702 Chair Alex Soupir Single Cell Spatial RNA and Protein in Cancer Research	SESSION 703 Chairs Riyue Bao Brent T. Schlegel Computational Frontiers in Patient-Derived Organoids
5:15 - 5:45 pm	Lighting talks #1	Lighting talks #2	Lighting talks #3

DAY 3 March 29, 2026 (Sunday)			
7:30 - 8:30 am	BREAKFAST		
7:30 - 8:30 am	MCBIOS Board Meeting (By Invitation Only) Patel Room		
	Ted & Marty Couch Auditorium	SRB Ferman Family Room	SRB David Murphey Room
8:30 - 9:45 am	SESSION 801 Chair Jamie Teer Electronic Notebooks for Data Science: Strategies, Suggestions, Experiences	SESSION 802 Chairs Tingting Hou Xiaoxi Shen Advances in Statistical and Machine Learning Models for Genetics and Genomics	SESSION 803 Chairs Aleksandra Karolak Katarzyna Mizgalska Marina Gavrilovskaia Introduction to Molecular Design: From Molecular Drawing to Computational Drug Design
9:45 – 10:00 am	Break		
10:00 - 11:15 am	SESSION 901 Chairs Sophia Song, Mingxiang Teng Statistical Learning and Applications in Multi-Omics	SESSION 902 Jincheng Shen Ann Chen AI and Data Science Innovations in Precision Oncology	SESSION 903 Chair Paul Stewart Computational mass spectrometry
11:15 - 11:30 am	Break		
11:30 - 12:00 pm	Announcement of journal partnership (Auditorium)		
12:00 - 12:30 pm	AWARDS and CLOSING CEREMONY		

MCBIOS 2026 Program

DAY 1 : Friday 3/27/26

7:30 am – 8:30 am	**REGISTRATION & BREAKFAST** **Ted & Marty Couch Auditorium & SRB Ferman Family Room & SRB David Murphey Room**
Welcome Remarks **Mingxiang Teng, Ph.D., Qianqian Song, Ph.D., Xuefeng Wang, Ph.D.** **MCBIOS 2026 Co-Chairs** **Michael Schell, Ph.D.** **Interim Division Chief of Quantitative Science, Moffitt Cancer Center** **Interim Chair of Biostatistics and Bioinformatics, Moffitt Cancer Center** 8:30 am – 8:45 am	
KEYNOTE 1: Rafael A. Irizarry, Ph.D. **Chair and Professor of Department of Data Science, Dana-Farber** **Title: Statistical Challenges in Spatial Transcriptomics** 8:45 am – 9:45 am	
Session 101: Innovative Computational Approaches for Decoding Biological Complexity from Single-Cell Data **Session Chair: Shenglin Mei**	
Room: Ted & Marty Couch Auditorium	
9:45 am – 11:00 am	1. Interpretable machine learning method to enable ultra-resolution analyses of whole transcriptome spatial data with high definition **Yi Zhang**, Duke University 2. Unbiased analytical methods of chromatin dynamics and transcriptional regulation **Sheng'en Shawn Hu**, University of Virginia 3. Computational Models of Metabolite-Mediated Intercellular Communications **Rongbin Zheng**, Boston Children's Hospital/Harvard 4. scSpaDA: Sparse discriminant analysis for decoding disease-associated cell populations and genes **Shenglin Mei**, Virginia Tech
Session 102: Multi-omics integration to decipher disease mechanisms **Session Chair: Jianrong Wang**	
Room: SRB Ferman Family Room	
9:45 am – 11:00 am	1. Computational modeling of metabolic changes in a body context **Chi Zhang**, Oregon Health and Science University 2. Modeling high resolution spatial transcriptional data without cell segmentation **Sha Cao**, Oregon Health and Science University 3. Interpret human genetic variants via single-cell multimodal omics and AI models **Yang Li**, Washington University at St Louis
Session 103: AI in Genomics	

Session Chair: Yan Guo	
Room: SRB David Murphey Room	
9:45 am – 11:00 am	1. Enhancing Spatial Transcriptomics Using Hybrid Transformer–Graph Convolutional Networks **Steven Yan**, University of Miami 2. CAPTATe: a novel AI-driven tool for enhanced artifact filtering and precise variant detection in single-cell DNA sequencing data **Luigi Ferraro**, University of Miami 3. Artificial Intelligence for Decoding Tumor Microenvironment and Drug Response **Qianqian Song**, University of Florida
11:00 am – 11:15 am	**BREAK**
Session 201: Computational Digital Twins for Translational Biomedical Informatics **Session Chair: Jing Su**	
Room: Ted & Marty Couch Auditorium	
11:15 am – 12:30 pm	1. TBD **Jing Su**, Indiana University School of Medicine 2. TBD **Jake Chen**, University of Alabama at Birmingham 3. TBD **Stephan Schürer**, University of Miami
Session 202: Next-Generation Web Visual Analytics for Bioinformatics and Artificial Intelligence **Session Chairs: Xijin Ge/Timothy Shaw**	
Room: SRB Ferman Family Room	
11:15 am – 12:30 pm	1. Datably: AI-powered bioinformatics platform **Xijin Ge**, South Dakota State University 2. Web tools for Drug Discovery **Yu-Chiao Chiu**, University of Pittsburgh 3. Web tools for multiomics integration (proteomics and metabolomics) **Paul Stewart**, University of Utah
Session 203: From Omics to Network Pharmacology and Systems Medicine **Session Chair: Zongliang Yue**	
Room: SRB David Murphey Room	
11:15 am – 12:30 pm	1. Integrative Spatial Transcriptomics and Network Biology Analysis Identifies Novel Drug Targets in Lung Adenocarcinoma **Ece (Gamsiz) Uzun**, Brown University 2. AI-driven single-cell genomics analysis decodes brain diseases from genes to network medicine **Daifeng Wang**, University of Wisconsin-Madison 3. A Protein Interactome-Based Framework Reveals the General Toxicity of Chemicals **Xiao Gan**, Nanjing University of Information Science & Technology 4. GOLDEN-GNN: Advancing Interpretable Cross-Domain Data Harmonization in the CFDE **Zongliang Yue**, Auburn University
12:30 pm – 12:40 pm	**Group Pictures**
12:40 pm – 1:30 pm	**LUNCH**

KEYNOTE 2: Ting Wang, Ph.D. **Chair and Professor of Department of Genetics, Washington University** **Title: The Human Pangenome Project** 1:30 pm – 2:30 pm	
Session 301: Adverse Event: A Gold Mine for Data Science in Cancer Research **Session Chairs: Dung-Tsa Chen/ Aik Choon Tan**	
Room: Ted & Marty Couch Auditorium	
2:30 pm – 3: 45 pm	1. GLP-1 Receptor Agonists (GLP-1RAs) and Cancer Outcomes **Rotana Radwan**, Purdue University 2. Walking the Tightrope: Balancing Immunotherapy Efficacy and Toxicity **Aik Choon Tan**, University of Utah 3. Hidden Treasure of Adverse Event Data in Cancer Clinical Trial **Dung-Tsa Chen**, Moffitt Cancer Center
Session 302: Adverse Event: Next-Generation Genomics and Bioinformatics for Pediatric Cancer and Congenital Disorders **Session Chairs: Jatinder Lamba/ Timothy I Shaw**	
Room: SRB Ferman Family Room	
2:30 pm – 3: 45 pm	1. Variants associated with drug resistance **Jatinder Lamba**, University of Florida 2. Therapy-related clonal hematopoiesis **Changde Cheng**, University of Alabama at Birmingham 3. Identifying genetic predisposition of congenital diseases and pediatric cancer **Wenan Chen**, Mayo Clinic 4. Accelerated Aging in Pediatric Cancer **Zhaoming Wang**, University of South Florida 5. Statistical methodology in pediatric cancer **Stanley Pounds**, St Jude Children's Research Hospital
Session 303: From Data to Discovery: Statistical Learning in Biomedicine **Session Chair: Hui Jiang**	
Room: SRB David Murphey Room	
2:30 pm – 3: 45 pm	1. Transforming healthcare using AI-powered wearables **Meng Wang**, University of Michigan 2. Incorporating Large Language Model Embeddings for Variation across the Human Genome **Didong Li**, UNC-Chapel Hil 3. Graph-Based Continuous Spatial Domains for Spatiotemporal Tissue Mapping in Spatial Transcriptomics **Wenpin Hou**, Columbia University 4. Boosting local biomedical data analysis with external information **Hui Jiang**, University of Michigan
3:45 pm - 4:00 pm	**BREAK**
Session 304: Single-Cell and Multi-Omics Genomics: Bridging Data and Innovation to Transform Health **Session Chair: Zhaotong Lin**	
Room: Ted & Marty Couch Auditorium	
4:00 pm – 5:15 pm	1. Integrative Lineage Inference and Multi-Omics Cell Clustering from Single-Cell DNA Sequencing

	Xian Mallory, Florida State University 2. High Similarity of cis Genetic Effect Sizes on Gene Expression Across Cell Types **Chang Su**, Emory University 3. Distilling Direct Effects via Conditional Differential Gene Expression Analysis **Jiaqi Gu**, University of South Florida 4. scTWAS: A powerful statistical framework for single-cell transcriptome-wide association studies **Zhaotong Lin**, Florida State University
Session 305: Student Flash Talks from the ISCB Regional Meeting 2025 **Session Chairs: Leon Hardy/Timothy I Shaw**	
Room: SRB Ferman Family Room	
4:00 pm – 5:15 pm	Student Flash Talks from the ISCB Regional Meeting 2025
Session 306: YSEA	
Room: SRB David Murphey Room	
4:00 pm – 5:15 pm	YSEA – Post-Doc Research Fellows YSEA – Undergraduate Students YSEA – Graduate Students
5:15 pm – 6:15 pm	**MCBIOS Board Member Voting (Auditorium)**
	RECEPTION, POSTER SESSION (SRB Rooms)
6:30 pm	**MCBIOS Business Session (By Invitation Only)**

DAY 2 : Saturday 3/28/26

7:30 am – 8:30am	**BREAKFAST** **Ted & Marty Couch Auditorium & SRB Ferman Family Room & SRB David Murphey Room**
Room: Ted & Marty Couch Auditorium	
KEYNOTE 3: Issam El Naqa, Ph.D. **Chair and Senior Member of Department of Machine Learning, Moffitt Cancer Center** **Title: AI in Cancer Research and Discovery: What did we learn so far?** 8:30 am – 9:30am	
9:30 am – 9:45am	**BREAK**
Session 401: Advanced Computational Methods in Single-cell and Spatial Omics **Session Chairs: Zhana Dunren/Xiuwei Zhang/Peng Qiu**	
Room: Ted & Marty Couch Auditorium	
9:45 am - 11:00am	1. PantheonOS: an evolvable AI agent platform for scientific discovery **Xiaojie Qiu**, Stanford University 2. Modeling and Predicting Single-cell Perturbation Responses **Wei Li**, University of Maryland School of Medicine 3. Graph signal processing is a bridge connecting spatial omics technologies and applications **Yuzhou Chang**, The Ohio State University 4. Scalable, Generalizable, and Uncertainty-Aware Integration of Spatial Multi-Omics Across Diverse Modalities and Platforms **Ying Ma**, Brown University

Session 402: Artificial Intelligence Innovations for Spatial Transcriptome Analysis **Session Chair: Li Chen**	
Room: SRB Ferman Family Room	
9:45 am - 11:00am	1. Image-based alignment of spatial omics data with AXIS **Zhicheng Ji,** Duke University 2. Winnow-KAN: Single Cell Location Recovery with Small-Gene-Set via Kolmogorov-Arnold Network **Qihuang Zhang**, McGill University 3. highSpaClone: Copy Number Variation Inference and Tumor Subclone Analysis for High-Resolution Spatial Transcriptomics **Ziyi Li**, MD Anderson Cancer Center
Session 403: (Panel) Careers in Bioinformatics: A Conversation with Industry Leaders **Session Chair: Timothy I Shaw**	
Room: SRB David Murphey Room	
9:45 am – 11:00 am	1. Career advice from an academic faculty in genomics **Rays Jiang**, University of South Florida 2. Building a career jamming with AI **Eric Talevich**, Owner Independent Consultant, Et al Bio LLC 3. Career as a biostatistics/bioinformatics manager **Richie Reich**, Moffitt Cancer Center
11:00 am – 11:15 am	BREAK
Session 501: AI in precision medicine **Session Chairs: Sophia Song/Mingxiang Teng**	
Room: Ted & Marty Couch Auditorium	
11:15 am – 12:30 pm	1. Harnessing big data for precision medicine **Leng Han**, Indiana University School of Medicine 2. TBD **Jeremy Goecks**, Moffitt Cancer Center 3. Teaching AI the grammar of resistance: genotype-to-phenotype prediction in the malaria parasite **Xiaoming Liu**, University of South Florida 4. Towards a foundation model for treatment-related adverse events in cancer immunotherapy **Xuefeng Wang**, Moffitt Cancer Center
Session 502: Novel machine learning methods in genetics and genomics **Session Chair: Zhaohui "Steve" Qin**	
Room: SRB Ferman Family Room	
11:15 am – 12:30 pm	1. Deep learning-based prognosis models accurately predict the time to delivery among preeclampsia patients using health records at the time of diagnosis **Lana Garmire**, University of Alabama at Birmingham 2. Data-driven selection of analysis decisions in single-cell RNA-seq trajectory inference **Rhonda Bacher**, University of Florida 3. mmSVG: A Scalable and Computationally Efficient Model-Based Framework for Detecting Spatially Variable Genes in Large-Scale Spatial Omics Data **Ximing Ran**, Emory University 4. SnapHiC-G: identifying long-range enhancer-promoter interactions from single cell Hi-C data via a global background model

	Ming Hu, Cleveland Clinic
Session 503: AI-Driven 3D Genome Reconstruction and Multimodal Learning: From Hi-C Enhancement and Forecasting to Multi-Omics and Single-Cell Modeling **Session Chair: Oluwatosin Oluwadare**	
Room: SRB David Murphey Room	
11:15 am – 12:30 pm	1. SO(3)-Equivariant Graph Neural Networks for 3D Chromosome Reconstruction from Single-Cell Hi-C Data **Jianlin Cheng**, University of Missouri, Columbia 2. Reconstruct single-cell 3D Genome Structures to understand context-specific Gene Regulation **Jianrong Wang**, Michigan State University 3. Identifying Chromatin jets and Higher-order Genome folding Structures **Minji Kim**, University of Michigan, Ann Arbor 4. Improving Hi-C Resolution through Chromatin Accessibility–Guided Deep Learning **Oluwatosin Oluwadare**, University of North Texas
12:30 pm – 1:30 pm	**LUNCH**
KEYNOTE 4: Jinghui Zhang, Ph.D. **Chair of Bioinformatics, St Jude Children's Research Hospital** **Title: Therapy-related mutagenesis in childhood cancer patients and survivors** 1:30 pm – 2:30 pm	
Session 601: Genomics and Translational Bioinformatics Working Group Session: AI Applications in Genomics and Bioinformatics **Session Chair: Ece Uzun**	
Room: Ted & Marty Couch Auditorium	
2:30 pm – 3:45 pm	1. Multimodal AI approaches to rare disease diagnosis incorporating clinical notes, omics and multimedia information **Kai Wang**, University of Pennsylvania 2. BrainGeneBot: A GPT-engineered platform for polygenic risk score analysis and interpretation in Alzheimer's disease **Zhongming Zhao**, University of Texas Health Science Center at Houston 3. HALO: delineating cellular states and dynamics from single cell multi-omics data **Takis Benos**, University of Pittsburgh and University of Florida 4. Predicting Glioma Recurrence Through Multimodal Attention-Based Deep Learning **Ece Uzun**, Brown University
Session 602: AI and Foundation Models in Biomedicine **Session Chair: Qianqian Song**	
Room: SRB Ferman Family Room	
2:30 pm – 3:45 pm	1. Ad hoc, post hoc, and intrinsic-hoc in AI and bioinformatics **Qin Ma**,Ohio State University 2. Noise-tolerant spatial multi-omics integration in semi-metric space **Junxin Wang**, Indiana University School of Medicine **3.** Foundation model for Bridging Pathology and Spatial Transcriptomics **Guangyu Wang**, Houston Methodist/Weill Cornell Medicine
Session 603: AI-Powered Bio-Discovery: A Multi-Level Collaborative Codeathon	

Session Chairs: Inimary Toby-Ogundeji/ Emil Jivishov	
Room: SRB David Murphey Room	
2:30 pm – 3:45 pm	1. Beyond the Classroom: Designing research Collaborative Codeathon/Hackathon Projects That Stick **Inimary Toby-Ogundeji**, University of Dallas 2. Peer-to-Peer Power: Why High Schoolers Listen to You More Than Their Teachers **Emil Jivishov** 3. Research Impact: Scaling Codeathon Projects for Social Good **Antonio Mestanza** 4. Trainee Codeathon Team Summaries (TBD)
3:45 pm – 4:00 pm	**BREAK**
Session 701: Machine Learning Advances in Single-cell Omics **Session Chairs: Peng Qiu/Xiuwei Zhang/Zhana Duren**	
Room: Ted & Marty Couch Auditorium	
4:00 pm – 5:15 pm	1. Graph Neural Networks and Prompt-Based Deep Learning for Single-Cell Data Analysis **Dong Xu**, University of South Florida 2. Deciphering upstream signaling pathways through interpreting cell state and niche correlations learned by graph transformer **Zuoheng Anita Wang**, Yale University 3. Deep Generative Models for Virtual Cells and Cellular Dynamics in Single-Cell and Spatial Omic **Jun Ding**, McGill University
Session 702: Single Cell Spatial RNA and Protein in Cancer Research **Session Chair: Alex Soupir**	
Room: SRB Ferman Family Room	
4:00 pm – 5:15 pm	1. TBD **Alex Soupir**, Moffitt Cancer Center 2. Determining functional niches from spatial transcriptomics experiments **Brooke Fridley**, Children's Mercy 3. TBD **Veera Baladandayuthapani**, University of Michigan 4. TBD **Chase Sakitis**, Children's Mercy
Session 703: Computational Frontiers in Patient-Derived Organoids **Session Chairs: Riyue Bao/ Brent T. Schlegel**	
Room: SRB David Murphey Room	
4:00 pm – 5:15 pm	1. Digital pathology in human specimens and PDOs **Riyue Bao**, University of Pittsburgh 2. 3D imaging/modeling on organoid metastasis **Keisuke Ishihara**, University of Pittsburgh 3. Drug response prediction of patient-derived organoids **Duy T Nguyen**, Moffitt Cancer Center 4. Preclinical drug development of patient-derived organoids and PDX **Erin George**, Moffitt Cancer Center
Lighting Talks #1 **Ted & Marty Couch Auditorium**	

5:15 pm – 5:45 pm	1. Quantitative Evaluation of Gene Function Hypotheses Using Large Language Model Embeddings **Yahao Tan**, UPMC Hilman Cancer Center & University of Pittsburgh 2. The Axis of Senescence and Epithelial-Mesenchymal Transition Underlies Pancreatic Cancer Progression **Xiaoying Wang**, Ohio State University 3. Transcript-Level NMD Classification **Thac Duong**, Moffitt Cancer Center 4. Modeling Complex Effects and Individual Variability in Multi-Paradigm fMRI with Nonlinear Mixed Models **Xiaoxuan Li**, Tulane University
Lighting Talks #2 **SRB Ferman Family Room**	
5:15 pm – 5:45 pm	1. Machine learning enables accurate prediction of cancer patient outcomes for immunotherapy using real-world clinical data **Alyssa Pybus**, Moffitt Cancer Center 2. Parcellation Scheme Guided Multi-scale Non-parametric Prototype Learning for Interpretable Brain MRI Classification **Ziyu Zhou**, Tulane University 3. A Multimodal Atlas of DNA Regulatory Elements Across Human and Mammalian Cells **Wanying Wu**, Washington University School of Medicine 4. scModCompare: Disentangling Shared Identity and Modality-Specific Regulatory Programs in Multimodal Single-Cell Data **Hechen Li**, Georgia Institute of Technology 5. TEvarSim: A comprehensive toolset for transposable element modeling and benchmarking **Jian Miao**, Texas Tech University Health Sciences Center
Lighting Talks #3 **SRB David Murphey Room**	
5:15 pm – 5:45 pm	1. Genomic Random Interval (GRIN2): A Comprehensive Framework to Identify Genomic Hot-spot Loci and Enable Multi-omics Integration. **Abdelrahman Elsayed**, St. Jude Children's Research Hospital 2. Spatiotemporal cell type deconvolution leveraging tissue structure **Macrina Lobo**, Georgia Institute of Technology 3. AI-Enhanced Aptamer Design: Addressing the Blood-Brain Barrier Challenge of Alzheimer's Disease Therapeutics **Christina He**, Alphamind Club, Carnegie Vanguard High School 4. Understanding cutaneous immune-related adverse events at single-cell resolution **Olivia J. Cheng**, Huntsman Cancer Institute of University of Utah 5. Quantitative Characterization of Cellular Plasticity via Spin-Glass Modeling **Ahmed Ghobashi**, Ohio State University

DAY 3 : Sunday 3/29/26

7:30 am – 8:30am	**BREAKFAST** **Ted & Marty Couch Auditorium & SRB Ferman Family Room & SRB David Murphey Room**
7:30 am – 8:30 am **Ballroom Lobby**	**MCBIOS Board Meeting (By Invitation Only) Patel Room**
Session 801: Electronic Notebooks for Data Science: Strategies, Suggestions, Experiences **Session Chair: Jamie Teer**	
Room: Ted & Marty Couch Auditorium	
8:30 am – 9:45am	**1. Jinpeng Liu**, University of Kentucky **2. Jamie K. Teer**, Moffitt Cancer Center **3. Isis Narvaez-Bandera**, Moffitt Cancer Center
Session 802: Advances in Statistical and Machine Learning Models for Genetics and Genomics **Session Chair: Tingting Hou/Xiaoxi Shen**	
Room: SRB Ferman Family Room	
8:30 am – 9:45am	1. A Unified Statistical Framework for Detecting Spatially Variable Genes in Spatial Transcriptomics **Tingting Hou**, Louisiana State University 2. Variable Importance Through Lazy Trained Neural Networks with Applications in Genetic Studies **Xiaoxi Shen**, Texas State University 3. Regularized Linear Modeling for Highly Correlated Single-Cell Features **Xiaoru Dong**, Texas A&M College of Dentistry 4. Neural Tangent Kernels for Nonlinear Genetic Risk Prediction **Heng Ge**, University of Florida
Session 803: (Tutorial) Introduction to Molecular Design: From Molecular Drawing to Computational Drug Design **Session Chair: Aleksandra Karolak/Katarzyna Mizgalska/Marina Gavrilovskaia**	
Room: SRB David Murphey Room	
8:30 am – 9:45am	1. **Aleksandra Karolak**, Moffitt Cancer Center 2. **Katarzyna Mizgalska**, Moffitt Cancer Center 3. **Marina Gavrilovskaia**, Moffitt Cancer Center
9:45 am – 10:00 am	**BREAK**
Session 901: Statistical Learning and Applications in Multi-Omics **Session Chairs: Sophia Song/Mingxiang Teng**	
Room: Ted & Marty Couch Auditorium	
10:00 am – 11:15 am	1. Bayesian bi-directional Mendelian randomization under correlated and uncorrelated pleiotropy using GWAS summary statistics **Siyi Chen**, Louisiana State University 2. Dissecting Tumor Heterogeneity and Evolution Using Single cell and Spatial Multiomic Profiling **Xiaofei Song**, Moffitt Cancer Center 3. Multi-dimensional data for treatment optimization **Yuan Liu**, Indiana University School of Medicine 4. Nonlinear embedding and integration of omics data: a fast and tuning-freeapproach **Tianwei Yu**, the Chinese University of Hong Kong - Shenzhen
Session 902: AI and Data Science Innovations in Precision Oncology **Session Chairs: Jincheng Shen/Ann Chen**	

Room: SRB Ferman Family Room	
10:00 am – 11:15 am	1. Leveraging Multimodal Data and LLMs for Accelerating Cancer Drug Repurposing **Ann Chen**, Univeristy of Utah 2. Missingness-Aware Deep Reinforcement Learning for Adaptive Treatment Strategies in Non–Small Cell Lung Cancer Using Real-World Data **Jincheng Shen**, Univeristy of Utah 3. Unraveling Causal Pathways: Bayesian Innovations for High-Dimensional Mediation in Omics Research **Xi Qiao**, University of Utah 4. Enhancing clinical trial safety monitoring through AI-assisted toxicity assessment **Gabby Fort**, University of Utah
Session 903: Computational mass spectrometry **Session Chair: Paul Stewart**	
Room: SRB David Murphey Room	
10:00 am – 11:15 am	1. Deciphering antigen presentation in the tumor microenvironment with immunopeptidomics **Alex M. Jaeger**, Moffitt Cancer Center 2. spatialMET: An open and scalable framework for spatial metabolomics domain detection and spatial pattern analysis **Paul A. Stewart**, University of Utah 3. Systems Metabolism: Characterizing the interface between signaling and metabolism in Obesity **Tigist Y Tamir**, University of North Carolina at Chapel Hill 4. Decoding PTM Signaling with AI: From Substrate Maps to Kinase Activity **Xusheng Wang**, University of Tennessee Health Science Center
11:15 am - 11:30 am	**BREAK**
11:30 am - 12:00 am	**Announcement of journal partnership (Auditorium)**
12:00 am - 12:30 pm	**AWARDS and CLOSING CEREMONY**